

BCSE309L - CRYPTOGRAPHY & NETWORK SECURITY

* Security - Protection from danger, harm or unwanted threats.

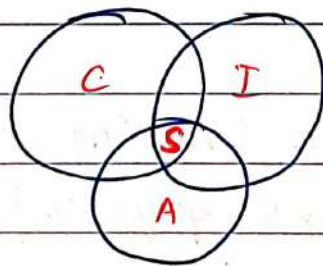
* Computer Security - Generic name for collection of tools designed to protect data.

Network Security - Protect data during transmission.

Internet Security - Protect data during transmission over collection of interconnected networks.

* Basic Security Components

CIA Triad.

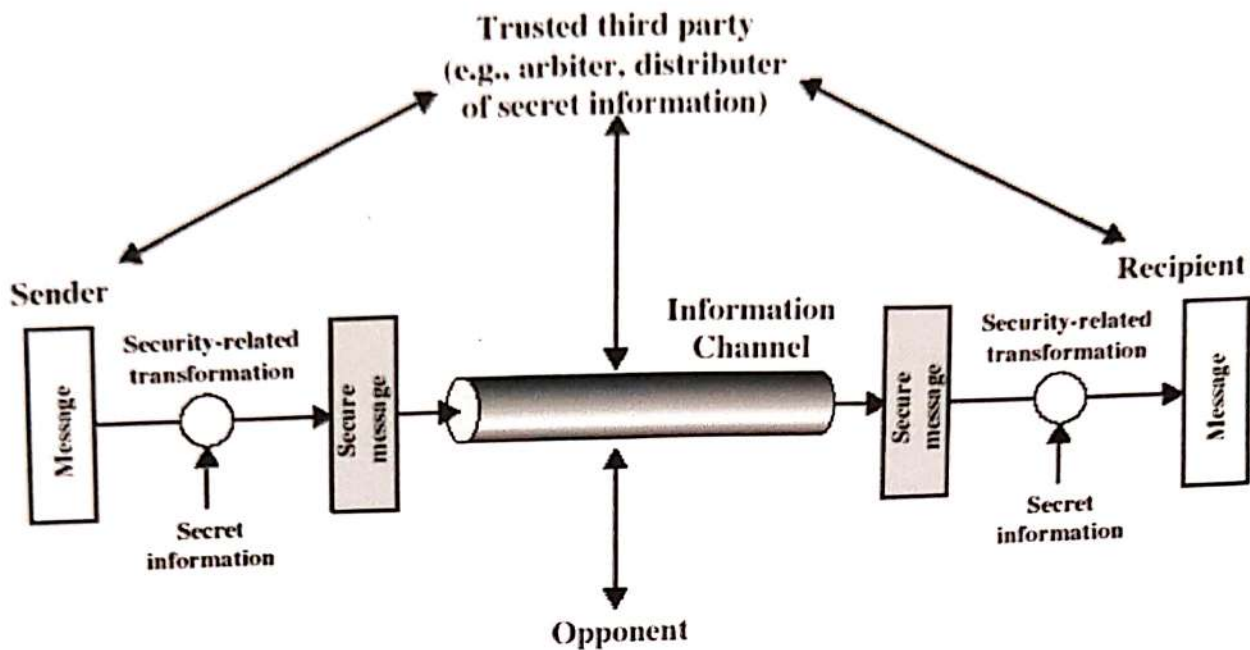


C - Confidentiality: The assets are accessible only by authorized parties.

I - Integrity: The assets are modified only by authorized parties & only in authorized ways.

A - Availability: Assets are accessible to authorized parties at all times.

* Model for network security.



* Cryptography - Designing secure systems.
 Cryptanalysis - Breaking or testing the cryptography systems.

Cryptology - Broader discipline involving both code-making (cryptography) and code breaking (cryptanalysis).

MODULE 1: FUNDAMENTALS OF NUMBER THEORY

I. FINITE FIELDS & NUMBER THEORY

* Finite field (Galois Field) is a mathematical structure with a finite number of elements where you can perform addition, subtraction, multiplication and division (except 0) and the results always stay within the set.

* Field is a set with 2 operations (addition & multiplication) that satisfies rules like associativity, commutativity, distributivity & existence of additive & multiplicative inverses.

* A finite field is simply a field with finite number of elements.

* Denoted as $GF(p^n)$ where

p - Prime number

n - +ve integer

The total number of elements is p^n .

* Examples

(i) $GF(2)$ - Simplest field with 2 elements $\{0, 1\}$

Addition = $1 + 1 = 0 \pmod{2}$

Multiplication = $1 \cdot 1 = 1 \pmod{2}$

(ii) $GF(5)$: A field with elements $\{0, 1, 2, 3, 4\}$ where operations are modulo 5.

(iii) $GF(7)$: A field with elements $\{0, 1, 2, 3, 4, 5, 6\}$ where operations are modulo 7.

* Additive & Multiplicative Inverses.

↳ Additive Inverse: For any element 'a' in a field, its additive inverse is element that, when added to a, gives additive identity 0.

$$a + (-a) = 0$$

Example: (i) Additive inverse of 7 is -7. [Integers]

(ii) In $GF(5)$: additive inverse of 3 is 2

$$\text{Since } 3 + 2 \equiv 0 \pmod{5}$$

↳ Multiplicative Inverse: For any non zero element a in a field, its multiplicative inverse is the element that, when multiplied with a, gives the multiplicative identity 1.

$$a \cdot a^{-1} = 1$$

Example: (i) In real numbers: The multiplicative inverse of 5 is $1/5$.

(ii) In $GF(7)$: the multiplicative inverse of 3 is 5, since $3 \cdot 5 \equiv 15 \equiv 1 \pmod{7}$.

$GF(2^n)$ Fields,

* When we work with computers, the positive integers are stored in the computer as n -bit words in which n is usually 8, 16, 32, 64 and so on.

* This means, the range of integers is 0 to $2^n - 1$.
The modulus is 2^n .

* We have 2 choices if we want to use a field,

(i) $GF(P^n)$

(ii) $GF(2^n)$.

* We can work in $GF(2^n)$ and use 2^n elements.
The elements in the set are n -bit words.

Example, if $n=3$, the set is

$$\{000, 001, 010, 011, 100, 101, 110, 111\}$$

* The n -bit words can be represented using polynomials.

Example: 8-bit word $\rightarrow 10011001$

$$\begin{array}{cccccccc} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \swarrow & \swarrow \\ 1x^7 + 0x^6 + 0x^5 + 1x^4 + 1x^3 + 0x^2 + 0x^1 + 1x^0 \\ = x^7 + x^4 + x^3 + 1 \end{array}$$

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* Similarly vice-versa can also be done.

Example: To find the 8-bit word related to the polynomial $x^5 + x^2 + x$, we first supply the omitted terms. Since $n=8$, it means the polynomial is of degree 7. The expanded polynomial is

$$0x^7 + 0x^6 + 1x^5 + 0x^4 + 0x^3 + 1x^2 + 1x^1 + 0x^0$$

This is related to 8 bit word 00100110.

* Addition of 2 polynomials never creates a polynomial out of the set. However, multiplication of 2 polynomials may create a polynomial with degree more than $n-1$. This means we need to divide the result by a modulus & keep only the remainder.

* The modulus in this case acts as a prime polynomial, which means no polynomials in the set can divide this polynomial.

* A PRIME POLYNOMIAL cannot be factored into a polynomial with degree of less than n . Such polynomials are referred to as IRREDUCIBLE POLYNOMIALS.

* List of irreducible polynomials of degree 1 to 5,

Degree Irreducible Polynomials

1	$(x+1), x$
2	x^2+x+1
3	$(x^3+x^2+1), (x^3+x+1)$
4	$(x^4+x^3+x^2+x+1), (x^4+x^3+1),$ (x^4+x+1)
5	$(x^5+x^2+1) (x^5+x^3+x^2+x+1),$ $(x^5+x^4+x^3+x+1), (x^5+x^4+x^3+x^2+1),$ $(x^5+x^4+x^2+x+1).$

* Polynomial addition

$$(x^5+x^2+x) \oplus (x^3+x^2+1) \text{ in } GF(2^8).$$

↳ polynomial addition.

$$\begin{array}{r}
 0x^7 + 0x^6 + 1x^5 + 0x^4 + 0x^3 + 1x^2 + 1x + 0x^0 \\
 0x^7 + 0x^6 + 0x^5 + 0x^4 + 1x^3 + 1x^2 + 0x + 1x^0 \quad \oplus \\
 \hline
 0x^7 + 0x^6 + 1x^5 + 0x^4 + 1x^3 + 0x^2 + 1x + 1x^0 \\
 \hline
 = x^5 + x^3 + x + 1.
 \end{array}$$

* Polynomial Multiplication.

Find the result of $(x^5+x^2+x) \otimes$

$(x^7+x^4+x^3+x^2+x)$ in $GF(2^8)$ with irreducible polynomial $(x^8+x^4+x^3+x+1).$

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$$P_1 \otimes P_2 = x^5(x^7 + x^4 + x^3 + x^2 + x) + x^2(x^7 + x^4 + x^3 + x^2 + x) + x(x^7 + x^4 + x^3 + x^2 + x)$$

$$= x^{12} + x^9 + x^8 + x^7 + x^6 + x^9 + x^6 + x^5 + x^4 + x^3 + x^8 + x^5 + x^4 + x^3 + x^2$$

$$= (x^{12} + x^7 + x^2) \bmod (x^8 + x^4 + x^3 + x + 1)$$

$$\begin{array}{r} x^4 + 1 \\ x^8 + x^4 + x^3 + x + 1 \overline{) x^{12} + x^7 + x^2} \\ \underline{x^{12} + x^8 + x^7 + x^5 + x^4} \\ x^8 + x^5 + x^4 + x^2 \\ \underline{x^8 + x^4 + x^3 + x + 1} \\ x^5 + x^3 + x^2 + x + 1 \end{array}$$

$$P_1 \otimes P_2 = x^5 + x^3 + x^2 + x + 1.$$

II . MODULAR ARITHMETIC

$$* a \bmod n = r.$$

$\downarrow$ $\downarrow$ $\rightarrow$ Residue [non-negative].
 Input modulus

* For any integer, $a = qn + r$, $0 \leq r < n$

$$q = \lfloor a/n \rfloor$$

$$a = \lfloor a/n \rfloor * n + (a \bmod n).$$

* Examples

$$11 \bmod 7 = 4$$

$$-11 \bmod 7 = -4 \Rightarrow -4 + 7 = 3.$$

$$27 \bmod 5 = 2$$

$$36 \bmod 12 = 0$$

$$-18 \bmod 14 = -4 \Rightarrow -4 + 14 = 10.$$

$$-7 \bmod 10 = -7 \Rightarrow -7 + 10 = 3.$$

Set of Residues : $\mathbb{Z}_n$.

* The result of modulo operation with modulus n is always an integer between 0 and $n-1$. In other words, the result of a $\bmod n$ is always a non-negative integer less than n .

* We can say that modulo operation creates a set, which in modular arithmetic is referred to as set of least residues modulo n (or) $\mathbb{Z}_n$.

* Examples,

$$\mathbb{Z}_n = \{0, 1, 2, 3, \dots, (n-1)\}$$

$$\mathbb{Z}_2 = \{0, 1\}, \mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$$

$$\mathbb{Z}_{11} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

Congruence

* In cryptography, we often use the concept of Congruence instead of equality.

* Mapping from $\mathbb{Z}$ to $\mathbb{Z}_n$ is not one to one. Infinite members of $\mathbb{Z}$ can map one member of $\mathbb{Z}_n$.

Ex: $2 \bmod 10 = 2, 12 \bmod 10 = 2.$

* To Show that two integers are congruent, we use congruence operator ($\equiv$).

Ex:

$$2 \equiv 12 \pmod{10}, 34 \equiv 24 \pmod{10}$$

$$-8 \equiv 12 \pmod{10}, -8 \equiv 2 \pmod{5}$$

Operations in $\mathbb{Z}_n$.

* The 3 binary operations - addition, subtraction, multiplication can be defined for set $\mathbb{Z}_n$.

* Operations

$$(a+b) \bmod n = c$$

$$(a-b) \bmod n = c$$

$$(a * b) \bmod n = c$$

* Examples

(i) Add 7 to 14 in $\mathbb{Z}_{15}$

$$(7+14) \bmod 15 = 6.$$

(ii) Subtract 11 from 7 in $\mathbb{Z}_{13}$.

$$(7-11) \bmod 13 = -4 \bmod 13 = 9.$$

(iii) Multiply 11 by 7 in $\mathbb{Z}_{20}$.

$$(7 * 11) \bmod 20 = 77 \bmod 20 = 17.$$

* Properties.

$$(i) (a+b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$$

$$(ii) (a-b) \bmod n = [(a \bmod n) - (b \bmod n)] \bmod n$$

$$(iii) (a * b) \bmod n = [(a \bmod n) * (b \bmod n)] \bmod n.$$

↳ Examples.

$$(i) (1723345 + 2124945) \bmod 11 = (8+9) \bmod 11 = 6.$$

$$(ii) (1723345 - 2124945) \bmod 11 = (8-9) \bmod 11 = 10$$

$$(iii) (1723345 * 2124945) \bmod 11 = (8 * 9) \bmod 11 = 6$$

$$* \text{ Also, } 10^n \bmod x = (10 \bmod x)^n \bmod x.$$

Relatively prime

* A set of elements are relatively prime to $\mathbb{Z}_n$ if their GCD is 1.

$$\mathbb{Z}_{10} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

$$\mathbb{Z}_{10}^* = \{1, 3, 7, 9\}$$

[Relatively prime]

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III. EUCLIDEAN ALGORITHM

* Finding GCD of 2 positive integers by listing all common divisors is not practical when integers are large.

* Euclidean algorithm helps in finding GCD for large numbers.

* Examples

(i) Find GCD (100, 35)

* Note: $\text{GCD}(a, 0) = a$.

When $b \neq 0$, $\text{GCD}(a, b) = \text{GCD}(|b|, a \bmod b)$

q	a	b	r
2	100	35	30
1	35	30	5
6	30	5	0
	GCD (5)	0	

(ii) GCD (2740, 1760)

q	a	b	r
1	2740	1760	980
1	1760	980	780
1	980	780	200
3	780	200	180
1	200	180	20
9	180	20	0
	GCD (20)	0	

EXTENDED EUCLIDEAN ALGORITHM

* For given integers a and b , the extended euclidean algorithm not only calculates the GCD d , but also two additional integers x and y that satisfy the following equation,

$$ax + by = d = \text{GCD}(a, b).$$

* Examples,

(i) Find $\text{GCD}(100, 35)$ & also find x, y that satisfies $ax + by = d$.

	q	x	y
100		1	0
35	2	0	1
30	1	1	-2
5	6	-1	3
0			

$$\text{GCD} = 5, x = -1, y = 3.$$

$$100(-1) + 35(3) = -100 + 105 = 5$$

$$\Downarrow$$

$$\text{GCD}(a, b).$$

(ii) 1759, 550

	q	x	y	
1759		1	0	
550	3	0	1	$1759(-111) +$
109	5	1	-3	$550(355)$
5	21	-5	16	$= -195249 +$
4	1	106	-339	$195250 = 1 \Rightarrow \text{GCD}$

$$\text{GCD} \Rightarrow 1$$

$$-111 \quad 355$$

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* x & y are multiplicative inverses of a & b respectively if $\text{GCD}(a, b)$ is 1.

* Practice problems

(i) Multiplicative inverse of 8 in $\mathbb{Z}_{10}$.

Ans: 8 does not have multiplicative inverse in $\mathbb{Z}_{10}$.

(ii) Find all multiplicative inverses in $\mathbb{Z}_{10}$.

	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9
2	0	2	4	6	8	0	2	4	6	8
3	0	3	6	9	2	5	8	1	4	7
4	0	4	8	2	6	0	4	8	2	6
5	0	5	0	5	0	5	0	5	0	5
6	0	6	2	8	4	0	6	2	8	4
7	0	7	4	1	8	0	2	9	6	3
8	0	8	6	4	2	0	8	6	4	2
9	0	9	8	7	6	5	4	3	2	1

Multiplicative inverse exists for $\{1, 3, 7, 9\}$

Multiplicative inverse of 1 is 1

" " 3 is 7

" " 7 is 3

" " 9 is 9.

(iii) Find multiplicative inverse of 11 in $\mathbb{Z}_{26}$.

	q	x	y	
26		1	0	
11	2	0	1	
4	2	1	-2	$26(3) + 11(-7)$
3	1	-2	5	$= 78 - 77 = 1$
1		3	-7	$\downarrow$ GCD

$$\begin{aligned}\text{Multiplicative inverse of } 11 \text{ in } \mathbb{Z}_{26} &= -7 \bmod 26 \\ &= -7 + 26 \\ &= \boxed{19}\end{aligned}$$

(iv) Find multiplicative inverse of 38 in $\mathbb{Z}_{180}$.

$$\text{GCD}(38, 180) = 2 \neq 1$$

$\therefore$ Multiplicative inverse does not exist for 38 in $\mathbb{Z}_{180}$.

* Fast Exponentiation.

$$(i) 17^{22} \bmod 21.$$

$$= (17^2)^{11} \bmod 21$$

$$= (17^2 \bmod 21)^{11} \bmod 21$$

$$= (16)^{11} \bmod 21$$

$$= (16^2)^5 * 16 \bmod 21$$

$$= (4)^5 * 16 \bmod 21 = \boxed{4}$$

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$$(ii) 35^{17} \mod 83$$

$$= (35^2)^{38} * 35 \mod 83$$

$$= (63)^{38} * 35 \mod 83$$

$$= (63^2)^{19} * 35 \mod 83$$

$$= (68)^{19} * 35 \mod 83$$

$$= (68^2)^9 * 68 * 35 \mod 83$$

$$= (59)^9 * 68 * 35 \mod 83$$

$$= (59^2)^4 * 59 * 68 * 35 \mod 83$$

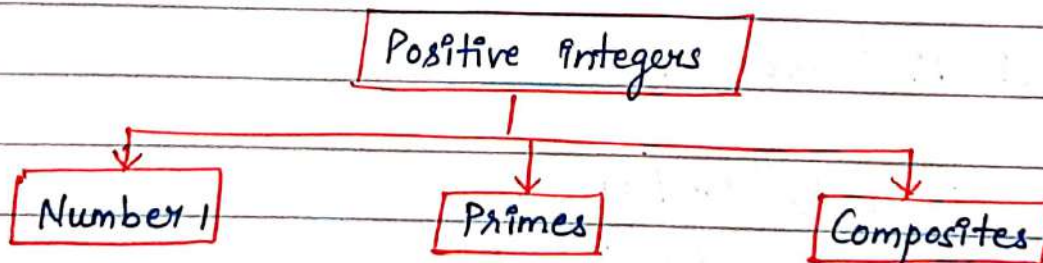
$$= (78)^4 * 67 \mod 83$$

$$= 44 * 67 \mod 83$$

$$= \boxed{43}$$

IV. PRIMES

* Positive integers can be divided into three groups:
The number 1, primes and composites.



* A positive integer is a prime iff it is exactly divisible by 2 integers, 1 and itself.

* A composite is a positive integer with more than 2 divisors.

* COPRIME - Two positive integers a and b , are relatively prime or coprime, if $\text{GCD}(a, b) = 1$.
The number 1 is relatively prime with any integer.

↳ If p is a prime, then all integers 1 to $p-1$ are relatively prime to p .

* CARDINALITY OF PRIMES
2 questions in primes,

(i) Is there a finite number of primes or is the list infinite? - ANS: Infinite no. of primes.

(ii) Given a number n , how many primes are smaller than or equal to n ? - ANS: NUMBER OF primes

⇒ INFINITE NUMBER OF PRIMES

↳ The number of primes is infinite.

↳ Informal proof: Suppose that the set of primes is finite, with p as the largest prime. Multiply the set of primes & call the result $P = 2 \times 3 \times \dots \times p$.
The integer $(P+1)$ cannot have factor $q \leq p$.
We know that q divides p . If q also divides P HQ

$(p+1)$, then q divides $(p+1) - p = 1$. The only number that divides 1 is 1, which is not a prime.

$\therefore q$ is larger than p .

Ex: Assume that only primes are in the set $\{2, 3, 5, 7, 11, 13, 17\}$. Here, $p = 510510$ & $p+1 = 510511$. However $510511 = 19 \times 97 \times 277$; none of these primes were in the original list. $\therefore$ There are 3 primes greater than 17.

$\Rightarrow$ NUMBER OF PRIMES.

* A function called $\pi(n)$ finds the number of primes smaller than or equal to n .

Ex: $\pi(1) = 0$, $\pi(2) = 1$, $\pi(3) = 2$, $\pi(10) = 4$,
 $\pi(20) = 8$, $\pi(50) = 15$, $\pi(100) = 25$.

* But, if n is large, how can we calculate $\pi(n)$?
 Ans: We can do by an approximation.

$$\left[\frac{n}{\ln n} \right] < \pi(n) < \left[\frac{n}{\ln n - 1.08366} \right].$$

Gauss discovered the upper limit. Lagrange discovered the lower limit.

* Example,

Find the no. of primes less than 1,00,000.

The approximation function gives range 72383 to 78543. The actual number of primes is 78498.

checking for primeness

* Given a number 'n', how can we determine if 'n' is a prime? The answer is that we need to see if the number is divisible by all primes less than $\sqrt{n}$.

* Example.

(i) Is 97 a prime?

Soln: The floor of $\sqrt{97} = 9$. The primes less than 9 are 2, 3, 5 and 7. We need to see if 97 is divisible by any of these numbers. It is not. So 97 is a prime.

(ii) Is 301 a prime?

Soln: The floor of $\sqrt{301}$ is 17. We need to check 2, 3, 5, 7, 11, 13 and 17. The numbers 2, 3 and 5 does not divide 301, but 7 does. $\therefore$ 301 is not a prime.

Sieve of Eratosthenes

* The Greek mathematician devised a method to find all primes less than n. This method is called SIEVE OF ERATOSTHENES.

* Suppose we want to find all primes less than 100.
 We write down all numbers between 2 & 100.
 Because $\sqrt{100} = 10$, we need to see if any number less than 100 is divisible by 2, 3, 5 and 7.

The following is the process.

(i) cross out all numbers divisible by 2 (except 2)

(ii) Similarly cross out all numbers divisible by 3, 5 & 7 (Except 3, 5 and 7).

(iii) The left over numbers are all primes.

~~2~~ ~~3~~ ~~4~~ ~~5~~ ~~6~~ ~~7~~ ~~8~~ ~~9~~ ~~10~~ ~~11~~ ~~12~~ ~~13~~ ~~14~~ ~~15~~
~~16~~ ~~17~~ ~~18~~ ~~19~~ ~~20~~ ~~21~~ ~~22~~ ~~23~~ ~~24~~ ~~25~~ ~~26~~ ~~27~~ ~~28~~
~~29~~ ~~30~~ ~~31~~ ~~32~~ ~~33~~ ~~34~~ ~~35~~ ~~36~~ ~~37~~ ~~38~~ ~~39~~ ~~40~~ ~~41~~
~~42~~ ~~43~~ ~~44~~ ~~45~~ ~~46~~ ~~47~~ ~~48~~ ~~49~~ ~~50~~ ~~51~~ ~~52~~ ~~53~~
~~54~~ ~~55~~ ~~56~~ ~~57~~ ~~58~~ ~~59~~ ~~60~~ ~~61~~ ~~62~~ ~~63~~ ~~64~~ ~~65~~
~~66~~ ~~67~~ ~~68~~ ~~69~~ ~~70~~ ~~71~~ ~~72~~ ~~73~~ ~~74~~ ~~75~~ ~~76~~ ~~77~~
~~78~~ ~~79~~ ~~80~~ ~~81~~ ~~82~~ ~~83~~ ~~84~~ ~~85~~ ~~86~~ ~~87~~ ~~88~~ ~~89~~
~~90~~ ~~91~~ ~~92~~ ~~93~~ ~~94~~ ~~95~~ ~~96~~ ~~97~~ ~~98~~ ~~99~~ ~~100~~.

Euler's phi function,

* Euler's phi function, $\phi(n)$, also called as EULER'S TOTIENT FUNCTION plays a very important role in cryptography.

* The function finds the number of integers that are both smaller than n and relatively prime to n .

* The function $\phi(n)$ calculates the number of elements in this set. The following helps to find the value of $\phi(n)$,

$$(i) \phi(1) = 0$$

$$(ii) \phi(p) = p - 1$$

$$(iii) \phi(m \times n) = \phi(m) * \phi(n) \text{ if } m \text{ and } n \text{ are relatively prime.}$$

$$(iv) \phi(p^e) = p^e - p^{e-1} \text{ if } p \text{ is a prime.}$$

* We can combine the above 4 rules to find the value of $\phi(n)$. Ex. if n can be factored as

$$n = p_1^{e_1} * p_2^{e_2} * \dots * p_k^{e_k}, \text{ then we can combine 3rd \& 4th rule to find}$$

$$\phi(n) = (p_1^{e_1} - p_1^{e_1-1}) * (p_2^{e_2} - p_2^{e_2-1}) * \dots * (p_k^{e_k} - p_k^{e_k-1}).$$

* It is very important to note that the value of $\phi(n)$ for large composites can be found only if the number n can be factored into primes.

* In other words, "The difficulty of finding $\phi(n)$ depends on the difficulty of finding the factorization of n ."

* Examples

(i) What is the value of $\phi(13)$?

Soln: 13 is prime.

$$\phi(13) = 13 - 1 = 12.$$

(ii) What is the value of $\phi(10)$?

Soln: Using rule 3

$$\begin{aligned}\phi(10) &= \phi(2) * \phi(5) \\ &= 1 * 4 \quad [\text{Rule 2}] \\ &= 4\end{aligned}$$

(iii) What is the value of $\phi(240)$?Soln: 240 can be written as $2^4 * 3^1 * 5^1$

$$\begin{aligned}\phi(240) &= (2^4 - 2^3) * (3^1 - 3^0) * (5^1 - 5^0) \\ &= 64\end{aligned}$$

(iv) Can we say that $\phi(49) = \phi(7) * \phi(7)$
 $= 6 * 6 = 36$?Soln: No. The third rule applies when m and n are relatively prime. Here $49 = 7^2$. We need to use the fourth rule:

$$\phi(49) = 7^2 - 7^1 = 42.$$

(v) What is the number of elements in Z_{14}^* ?Soln: The answer is $\phi(14) = \phi(7) * \phi(2) = 6 * 1 = 6$

The members are 1, 3, 5, 11 and 13.

V. FERMAT'S LITTLE THEOREM

* There are two versions of this theorem.

FIRST VERSION : The first version says that if 'p' is a prime and 'a' is an integer such that 'p' does not divide 'a', then $a^{p-1} \equiv 1 \pmod{p}$.

SECOND VERSION : The second version removes the condition on 'a'. It says that if 'p' is a prime and 'a' is an integer, then $a^p \equiv a \pmod{p}$.

* Fermat's little theorem sometimes is helpful for quickly finding a solution to some exponentiations.

* Examples

(i) Find the result of $6^{10} \pmod{11}$.

Soln: $6^{10} \pmod{11} = 1$. This is by first version of Fermat's little theorem where $p=11$.

(ii) Find the result of $3^{12} \pmod{11}$.

Soln:

$$3^{12} \pmod{11} = (3^{11}) * 3 \pmod{11}$$

$$= 3^{11} \pmod{11} * 3 \pmod{11}$$

$$= 3 \pmod{11} * 3 \pmod{11}$$

$$= 9 \pmod{11}$$

$$= 9.$$

Multiplicative Inverses

* A very interesting application of Fermat's theorem is in finding some multiplicative inverses quickly if modulus is prime.

* If 'p' is a prime and 'a' is an integer such that 'p' does not divide 'a', then,

$$a^{-1} \bmod p = a^{p-2} \bmod p.$$

proof: Multiply both sides by a.

$$a * a^{-1} \bmod p = a * a^{p-2} \bmod p$$

$$\Rightarrow a^{p-1} = 1 \bmod p \text{ [First version of Fermat's little theorem].}$$

* Examples,

$$\begin{aligned} \text{(i)} \quad 8^{-1} \bmod 17 &= 8^{17-2} \bmod 17 = 8^{15} \bmod 17 \\ &= 15 \bmod 17 = 15 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad 5^{-1} \bmod 23 &= 5^{23-2} \bmod 23 = 5^{21} \bmod 23 \\ &= 14 \bmod 23 = 14 \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad 60^{-1} \bmod 101 &= 60^{101-2} \bmod 101 = 60^{99} \bmod 101 \\ &= 32 \bmod 101 = 32 \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad 22^{-1} \bmod 211 &= 22^{211-2} \bmod 211 = 22^{209} \bmod 211 \\ &= 48 \bmod 211 = 48 \end{aligned}$$

VI. EULER'S THEOREM

* Euler's theorem can be thought of as generalization of Fermat's little theorem. The modulus in the Fermat theorem is a prime, the modulus in Euler's theorem is an integer.

* Two versions of this theorem are,

FIRST VERSION: The first version of Euler's theorem is similar to the first version of Fermat's little theorem. If a and n are coprime, then

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

SECOND VERSION: The second version of Euler's theorem is similar to second version of Fermat's little theorem; it removes the condition that a and n should be coprime. If $n = p \times q$, $a < n$ and k an integer, then

$$a^{k * \phi(n) + 1} \equiv a \pmod{n}.$$

* Examples

(i) Find the result of $6^{24} \pmod{35}$.

$$6^{\phi(35)} \pmod{35} = 1.$$

$$[\phi(35) = \phi(5) * \phi(7) \\ = 4 * 6 = 24]$$

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(ii) Find the result of $20^{62} \bmod 77$.

If we take $k=1$ on the second version, we have

$$\begin{aligned} 20^{62} \bmod 77 &= (20 \bmod 77) (20^{\phi(77)+1} \bmod 77) \bmod 77 \\ &= 20 * 20 \bmod 77 \\ &= 15. \end{aligned}$$

Multiplicative Inverses,

* Euler's theorem can be used to find multiplicative inverses modulo a prime; Euler's theorem can be used to find multiplicative inverses modulo a composite. If 'n' and 'a' are coprime, then

$$a^{-1} \bmod n = a^{\phi(n)-1} \bmod n.$$

* Examples,

$$\begin{aligned} \text{(i)} \quad 8^{-1} \bmod 77 &= 8^{\phi(77)-1} \bmod 77 = 8^{59} \bmod 77 \\ &= 29 \bmod 77 = 29 \end{aligned}$$

$$\text{(ii)} \quad 7^{-1} \bmod 15 = 7^{\phi(15)-1} \bmod 15 = 7^7 \bmod 15 = 13.$$

$$\begin{aligned} \text{(iii)} \quad 60^{-1} \bmod 187 &= 60^{\phi(187)-1} \bmod 187 = 60^{159} \bmod 187 \\ &= 53 \bmod 187 \\ &= 53 \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad 71^{-1} \bmod 100 &= 71^{\phi(100)-1} \bmod 100 = 71^{39} \bmod 100 \\ &= 31 \bmod 100 = 31. \end{aligned}$$

VII. CHINESE REMINDER THEOREM

* The Chinese Remainder Theorem (CRT) is used to solve a set of congruent equations with one variable but different moduli, which are relatively prime.

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

...

$$x \equiv a_k \pmod{m_k}$$

* The CRT states that above equations have a unique solution if the moduli are relatively prime.

* The solution steps are as follows,

(i) Find $M = m_1 \times m_2 \times \dots \times m_k$. This is the common modulus.

(ii) Find $M_1 = M/m_1, M_2 = M/m_2, \dots, M_k = M/m_k$.

(iii) Find the multiplicative inverse of $M_1, M_2, \dots, M_k$ using the corresponding moduli $(m_1, m_2, \dots, m_k)$. Call the inverses $M_1^{-1}, M_2^{-1}, \dots, M_k^{-1}$.

(iv) The solution to the simultaneous equations is

$$x = (a_1 \times M_1 \times M_1^{-1} + a_2 \times M_2 \times M_2^{-1} + \dots + a_k \times M_k \times M_k^{-1}) \pmod{M}.$$

* Examples,

$$(1) \quad x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

Soln:

Given.

$$\hookrightarrow M = 3 * 5 * 7 = 105, \quad a_1 = 2, a_2 = 3, a_3 = 2$$

$$\hookrightarrow M_1 = M/m_1 = 105/3 = 35$$

$$m_1 = 3, m_2 = 5,$$

$$M_2 = M/m_2 = 105/5 = 21$$

$$m_3 = 7$$

$$M_3 = M/m_3 = 105/7 = 15$$

$$\hookrightarrow M_1^{-1} \pmod{m_1} = 35^{-1} \pmod{3} = 2$$

$$M_2^{-1} \pmod{m_2} = 21^{-1} \pmod{5} = 1$$

$$M_3^{-1} \pmod{m_3} = 15^{-1} \pmod{7} = 1$$

$$\hookrightarrow x = [(a_1 * M_1 * M_1^{-1}) + (a_2 * M_2 * M_2^{-1}) + (a_3 * M_3 * M_3^{-1})] \pmod{M}$$

$$= [(2 * 35 * 2) + (3 * 21 * 1) + (2 * 15 * 1)] \pmod{105}$$

$$= 23 \pmod{105}$$

$$\boxed{x = 23}$$

$$(2) \quad x \equiv 3 \pmod{7}$$

$$x \equiv 3 \pmod{13}$$

$$x \equiv 0 \pmod{12}$$

Given

$$\text{Soln: } M = 7 * 13 * 12 = 1092, \quad a_1 = 3, a_2 = 3, a_3 = 0$$

$$m_1 = 7, m_2 = 13, m_3 = 12$$

$$M_1 = M/m_1 = 1092/7 = 156$$

$$M_2 = M/m_2 = 1092/13 = 84$$

$$M_3 = M/m_3 = 1092/12 = 91$$

$$\begin{aligned} M_1^{-1} \bmod m_1 &= 156^{-1} \bmod 7 \\ &= 156^{7-2} \bmod 7 \quad [\text{Fermat's theorem}] \\ &= 156^5 \bmod 7 \\ &= (156^2)^2 * 156 \bmod 7 \\ &= 4^2 * 156 \bmod 7 \\ &= 4 \end{aligned}$$

$$\begin{aligned} M_2^{-1} \bmod m_2 &= 84^{-1} \bmod 13 \\ &= (84 \bmod 13)^{-1} \bmod 13 \\ &= 6^{-1} \bmod 13 \\ &= 6^{13-2} \bmod 13 \quad [\text{Fermat's theorem}] \\ &= 6^{11} \bmod 13 \\ &= 11 \end{aligned}$$

$$x = ([a_1 * M_1 * M_1^{-1}] + [a_2 * M_2 * M_2^{-1}] + [a_3 * M_3 * M_3^{-1}]) \bmod M$$

$[a_3 \text{ is } 0, \text{ so the full term is } 0]$

$$= [(3 * 156 * 4) + (3 * 84 * 11)] \bmod 1092$$

$$= (1872 + 2772) \bmod 1092$$

$$x = 276$$

VIII. DISCRETE LOGARITHMS

* Discrete logarithms are fundamental to asymmetric key cryptography.

The Powers of an Integer, Modulo n .

* Consider an operation mod 5.

Take all powers of an integer 'a' mod 5, for all positive $a < 5$.

a	a^2	a^3	a^4	(mod 5)
1	1	1	1	
2	4	3	1	
3	4	2	1	
4	1	4	1	

* The values generated in each row forms a sequence.

* Length of sequences end in 1.

Ex.: For 1, length of sequence is 1.

For 2, length of sequence is 4.

For 3, length of sequence is 4.

For 4, length of sequence is 2.

* Length of sequence divides $\phi(p)$.

$$\phi(5) = 5 - 1 = 4.$$

The previously found sequence lengths divide 4.

* Cyclic group - It is a group that can generate all elements of the group.

Here, the group $\mathbb{Z}_5$ is cyclic, generated by 2 and 3 (All elements of $\mathbb{Z}_5$ are generated by integers 2 & 3 raised to powers).

The elements 2 & 3 here are called as GENERATORS (or) PRIMITIVE ROOTS.

Logarithms of Modular Arithmetic

* With ordinary positive real numbers, the logarithm function is the inverse of exponentiation. A similar function exists for modular arithmetic.

* A BRIEF REVIEW OF ORDINARY LOGARITHM PROPERTIES

↳ The logarithm of a number is defined to be the power to which some positive base (except 1) must be raised in order to equal the number. That is, for base x & for a value y ,

$$y = x^{\log_x(y)}$$

↳ The properties of logarithm include

$$(i) \log_x(1) = 0 \quad (ii) \log_x(x) = 1$$

$$(iii) \log_x(yz) = \log_x(y) + \log_x(z)$$

$$(iv) \log_x(y^a) = a * \log_x(y)$$

HQ

* Consider a primitive root 'a' for some prime number 'p'. Then we know that the powers of 'a' from 1 through (p-1) produce each integer from 1 through (p-1) exactly once.

* We also know that any integer 'b' satisfies

$$b \equiv r \pmod{p} \text{ for some } r, \text{ where } 0 \leq r \leq (p-1)$$

[by the definition of modular arithmetic]

It follows that for any integer 'b' & a primitive root 'a' of prime number 'p', we can find unique exponent 'i' such that

$$b \equiv a^i \pmod{p} \text{ where } 0 \leq i \leq (p-1)$$

This exponent 'i' is referred to as the discrete logarithm of the number 'b' for the base a (mod p). We denote this as,

$$\text{dlog}_{a,p}(b)$$

* Note: $\text{dlog}_{a,p}(1) = 0$ because $a^0 \bmod p = 1 \bmod p = 1$.

and $\text{dlog}_{a,p}(a) = 1$ because $a^1 \bmod p = a$.

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\log_{2,10}(a)$	18	1	13	2	16	14	6	3	8	17	12	15	5	7	11	4	10	9

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\log_{3,10}(a)$	18	7	1	14	4	8	6	3	2	11	12	15	17	13	5	10	16	9

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\log_{10,19}(a)$	18	17	5	16	2	4	12	15	10	1	6	3	13	11	7	14	8	9

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\log_{13,19}(a)$	18	11	17	4	14	10	12	15	16	7	6	3	1	5	13	8	2	9

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\log_{14,19}(a)$	18	13	7	8	10	2	6	3	14	5	12	15	11	1	17	16	4	9

a	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
$\log_{15,19}(a)$	18	5	11	10	8	16	12	15	4	13	6	3	7	17	1	2	14	9